Equations and formulae — One-step · Two-step · Letters on both sides · Formulae

No calculator. Show the balancing step on each line, and check every answer. ★ = challenge question.

A · ONE-STEP AND TWO-STEP EQUATIONS

1 Solve these equations.

a $x - 7 = 12$ **b** $4x = 52$ **c** $\frac{x}{6} = 3$ **d** $x + 9 = 4$

> **a** Add 7 to both sides: $x = 19$. **b** Divide both sides by 4: $x = 13$. **c** Multiply both sides by 6: $x = 18$. **d** Subtract 9: $x = -5$ (check: $-5 + 9 = 4$ ✓).

2 Solve these equations.

a $3x - 4 = 20$ **b** $\frac{x}{5} + 2 = 6$ **c** $25 - 4x = 5$ **d** $4(x + 3) = 36$

e $\frac{2x}{3} = -8$

> **a** $+4$ then $\div 3$: $3x = 24$, $x = 8$. **b** -2 then $\times 5$: $\frac{x}{5} = 4$, $x = 20$. **c** $+4x$ to both sides: $25 = 5 + 4x$, so $4x = 20$, $x = 5$. **d** $\div 4$ first: $x + 3 = 9$, $x = 6$. **e** $\times 3$: $2x = -24$, $x = -12$.

3 Solve $\frac{2x + 7}{3} = 5$

> Multiply both sides by 3: $2x + 7 = 15$. Subtract 7: $2x = 8$, so $x = 4$. Check: $(8 + 7) \div 3 = 5$ ✓

B · EQUATIONS FROM ANGLES AND SHAPES

4 Three angles on a straight line are $5x$, $3x$ and x. Work out the size of each angle.

> Angles on a straight line add to $180°$: $5x + 3x + x = 180 \Rightarrow 9x = 180 \Rightarrow x = 20$. The angles are $100°$, $60°$ and $20°$.

5 The angles of a triangle are x, $2x + 15$ and $3x - 9$ (in degrees). Work out the size of each angle.

> Angles in a triangle add to $180°$:
> $x + 2x + 15 + 3x - 9 = 180 \Rightarrow 6x + 6 = 180 \Rightarrow x = 29$. The angles are $29°$, $73°$ and $78°$ (sum 180 ✓).

6 Two rectangles have the same area. One is 5 cm by $(x + 2)$ cm; the other is 3 cm by $(x + 6)$ cm. Work out the value of x.

> Equal areas: $5(x + 2) = 3(x + 6) \Rightarrow 5x + 10 = 3x + 18 \Rightarrow 2x = 8 \Rightarrow x = 4$.
> Check: $5 \times 6 = 30$ and $3 \times 10 = 30$ ✓

C · LETTERS ON BOTH SIDES

7 Solve these equations.

a $7m - 3 = 4m + 12$

b $4(n - 1) = 2(n + 3)$

c $3(x + 2) = 5(x - 2)$

> **a** Subtract $4m$ (the smaller m term): $3m - 3 = 12 \Rightarrow 3m = 15 \Rightarrow m = 5$.
> **b** Expand: $4n - 4 = 2n + 6 \Rightarrow 2n = 10 \Rightarrow n = 5$.
> **c** Expand: $3x + 6 = 5x - 10$. Subtract $3x$: $6 = 2x - 10 \Rightarrow 2x = 16 \Rightarrow x = 8$.

8 Leo says: "I think of a number, multiply it by 4 and subtract 7. If instead I multiply it by 2 and add 9, I get the same answer." Write an equation and find Leo's number.

> Number n: $4n - 7 = 2n + 9$. Subtract $2n$: $2n - 7 = 9 \Rightarrow 2n = 16 \Rightarrow n = 8$. Check: $32 - 7 = 25$ and $16 + 9 = 25$ ✓

D · WORKING WITH FORMULAE

9 The formula $v = u + at$ gives the speed v of an object after t seconds. Work out a when $u = 4$, $t = 5$ and $v = 29$.

> Substitute: $29 = 4 + 5a$. Subtract 4: $5a = 25$, so $a = 5$.

10 **a** Write a formula for converting a time in minutes (m) into hours (h).

b Use your formula to convert 390 minutes into hours.

> **a** There are 60 minutes in an hour, so $h = \dfrac{m}{60}$. **b** $h = \dfrac{390}{60} = 6.5$ hours.

11 The perimeter of a rectangle is $P = 2(l + w)$.

a Work out P when $l = 7.5$ cm and $w = 4$ cm.

b A rectangle has perimeter 30 cm and width 6 cm. Work out its length.

> **a** $P = 2(7.5 + 4) = 2 \times 11.5 = 23$ cm. **b** $30 = 2(l + 6) \Rightarrow l + 6 = 15 \Rightarrow l = 9$ cm.

12 $3x + 2y = 24$. If x and y are both positive integers, give all the possible pairs of values of x and y.

> $2y$ is even, so $3x$ must be even → x is even. $x = 2$: $2y = 18$, $y = 9$. $x = 4$: $2y = 12$, $y = 6$. $x = 6$: $2y = 6$, $y = 3$. $x = 8$ gives $y = 0$, not positive. $(2, 9), (4, 6), (6, 3)$.

13 A square of side x cm is cut from one corner of a rectangle 10 cm by 8 cm. The area of the shape that is left is 71 cm². Work out x.

> Area left = rectangle − square: $80 - x^2 = 71 \Rightarrow x^2 = 9 \Rightarrow x = 3$ cm (a length must be positive, so not -3).

Check: whatever you do to one side, do to the other · subtract the *smaller* letter term first · angles on a line 180°, in a triangle 180° · substitute back.